

TryHackMe — Computer Fundamentals Module

Inside a Computer

Learning Notes — Hardware Components & The Boot Process

"Trying to defend what you don't understand is like defending a castle you have never seen. In this room, we explore our castle."



1. Why Computer Hardware Matters for Security

Before you can protect a system, you must first understand what that system is made of. The TryHackMe 'Inside a Computer' room opens the Computer Fundamentals module with a castle analogy: just as a soldier must understand a castle's layout before defending it, a security professional must understand a computer's architecture before protecting it.

The Castle Analogy

Before protecting a castle, you need to know its layout: where the treasure room is, the food storage, the commander's quarters. You need to know who enters and how they enter. The bottom line: trying to defend what you don't understand is like defending a castle you have never seen. A computer is our castle — and this room is the tour.

1.1 The Human Body Analogy

The room uses a powerful and memorable analogy: every computer component has a human body equivalent. This makes abstract hardware concepts concrete and easy to remember:

Computer Component	Human Body Equivalent	Why the Analogy Works
Motherboard	Skeleton & Nervous System	Provides the physical structure and connects every other component — signals travel through it like nerves.

CPU	Brain	Constantly executes instructions — addition, logic, control — just as the brain processes thoughts and directs actions.
RAM	Short-term / Working Memory	Holds information temporarily while actively being used. Fast to access, but gone when power is removed.
Storage (SSD/HDD)	Long-term Memory	Retains data permanently — files, OS, programs — like memories stored even when not actively thinking.
Network Adapter	Vocal Cords	Enables communication with the outside world — sending and receiving data like speaking and listening.
PSU	Heart	Pumps power to every component — without it, nothing works, just as the heart pumps blood to every organ.
Graphics Card	Visual Cortex	Processes visual information from the OS and outputs it to a monitor, as the visual cortex processes sight.
I/O Devices	Senses (Eyes, Hands, Ears)	Input devices receive information (keyboard, mouse, microphone); output devices communicate back (monitor, speakers).

2. The Hardware Components — Deep Dive

Nearly every computer system — from a smartphone to a supercomputer — is built from the same fundamental building blocks. Each component has a specific role, and together they form a complete system capable of running software and providing services.

Motherboard ≈ Body: Skeleton & Nervous System

The motherboard is the central circuit board that physically connects and enables communication between all other components. It is the backbone of the computer.

- Every other component plugs into or connects via the motherboard — CPU, RAM, storage, GPU, network card, PSU.
- Contains the UEFI/BIOS chip — the firmware that controls the boot process before the OS loads.
- Houses the CPU socket, RAM slots (DIMMs), PCIe slots for expansion cards, SATA ports for storage.
- The chipset on the motherboard manages communication between the CPU, RAM, and peripherals.
- Form factors: ATX (full-size desktop), Micro-ATX, Mini-ITX (small form factor), ITX (ultra-compact).
- The motherboard determines CPU compatibility — each supports specific socket types (e.g. LGA1700 for Intel, AM5 for AMD).

Connects via: Directly — all components physically connect to it via sockets, slots, and headers.

Security relevance: Physical access to the motherboard allows resetting BIOS passwords, extracting RAM for cold-boot attacks, and modifying boot order to boot from USB. Hardware implants can be installed at the motherboard level.

CPU — Central Processing Unit ≈ Body: The Brain

The CPU is the computer's processor — it continuously fetches, decodes, and executes instructions from programs. Everything a computer does ultimately comes down to the CPU performing billions of operations per second.

- Modern CPUs have multiple cores — each core can execute instructions independently and in parallel.
- Clock speed (measured in GHz) indicates how many instruction cycles per second — higher = faster.
- CPU cache (L1, L2, L3) is ultra-fast on-chip memory that stores frequently used data to avoid slower RAM access.
- The CPU connects to the motherboard via the CPU socket — Intel uses LGA (Land Grid Array), AMD uses PGA (Pin Grid Array) or AM5.
- CPUs have two main architectures: x86-64 (Intel/AMD — most desktops/laptops) and ARM (most smartphones, Apple Silicon Macs).
- Thermal management: CPUs generate significant heat and require heatsinks and fans (or liquid cooling) to prevent throttling or damage.

Connects via: CPU socket on the motherboard — Intel LGA1700, AMD AM5.

Security relevance: CPU vulnerabilities include Spectre and Meltdown — speculative execution side-channel attacks that allow reading protected memory. These require OS/microcode patches. CPU-level rootkits (SMM rootkits) operate below the OS and survive reboots.

RAM — Random Access Memory ≈ Body: Short-term / Working Memory

RAM is the computer's volatile working memory — it holds the data and instructions the CPU is currently using. It is much faster than storage but loses all contents when power is removed.

- Volatile memory: all contents are erased when the computer powers off or restarts.
- Stores the running OS kernel, currently open applications, and active data being processed.
- Speed measured in MHz/GHz and data transfer rate (e.g. DDR5-6400 = 6400 MT/s).
- Modern systems use DDR4 or DDR5 (Double Data Rate) — DDR5 offers higher speed and lower power consumption.
- RAM modules (DIMMs) plug into dedicated slots on the motherboard — typically 2 or 4 slots.
- More RAM allows more programs to run simultaneously without the OS resorting to slower swap/virtual memory on disk.
- Capacity: modern systems typically have 8GB–64GB; servers may have hundreds of GB.

Connects via: DIMM slots on the motherboard.

Security relevance: Cold boot attacks: RAM contents persist briefly after power loss — by rapidly chilling RAM and transplanting it, an attacker can extract encryption keys (e.g. full-disk encryption keys) stored in RAM. RAM is also where malware resides while running — memory forensics analyses RAM to detect malicious processes.

Storage — SSD & HDD ≈ Body: Long-term Memory

Storage devices retain data permanently — even without power. The operating system, installed programs, and user files all live on storage. There are two primary types: traditional HDDs and modern SSDs.

- HDD (Hard Disk Drive): uses spinning magnetic platters and a read/write head. Mechanical movement limits speed. Still popular for large capacity at low cost (\$/GB). Prone to physical failure from drops.
- SSD (Solid State Drive): uses NAND flash memory chips — no moving parts. Much faster read/write speeds than HDD. Lower latency, more durable, silent. Higher cost per GB but falling rapidly.
- NVMe SSDs: connect via PCIe (M.2 slot) instead of SATA — 5–10x faster than SATA SSDs.
- SATA interface: connects HDDs and 2.5" SSDs to the motherboard via SATA data and power cables.
- Storage retains data even when powered off — unlike RAM.
- OS boot drive: the computer boots from the storage device containing the OS installer (bootloader).

Connects via: SATA cables (HDD/SATA SSD) or PCIe/M.2 slot (NVMe SSD).

Security relevance: Storage is the primary target for data theft, ransomware encryption, and forensic investigation. Deleted files often remain recoverable until overwritten. Full-disk encryption (BitLocker, VeraCrypt) protects data at rest. Secure erasure requires multiple overwrites or physical destruction for HDDs; SSDs require manufacturer-level secure erase commands.

Network Adapter ≈ Body: Vocal Cords

The network adapter (NIC — Network Interface Card) enables the computer to communicate with other devices over a network. It translates the computer's digital data into signals that can travel over a network medium.

- Wired NIC: connects via Ethernet cable (RJ45 port). Fast and reliable — typically 1Gbps or 2.5Gbps on consumer hardware.
- Wireless NIC (Wi-Fi card): communicates via radio waves — 802.11 standards (Wi-Fi 5/6/6E/7). Convenient but subject to interference and lower security.
- Most modern motherboards have both wired Ethernet and Wi-Fi built in (integrated NICs).
- Standalone NICs can be added via PCIe slots for higher performance or additional interfaces.
- Each NIC has a unique MAC address assigned by the manufacturer — used for Layer 2 (local network) communication.
- The NIC handles conversion between digital data (bits) and the physical signal — electrical, optical, or radio.

Connects via: Integrated on motherboard, or PCIe expansion slot.

Security relevance: The NIC is the entry point for all network-based attacks — malware C2 communications, lateral movement, exfiltration. MAC addresses can be spoofed. Promiscuous mode allows capturing all traffic on a segment (used by Wireshark/network monitoring). Rogue NICs or implanted hardware can silently transmit data.

♥ PSU — Power Supply Unit ≈ Body: The Heart

The PSU converts AC mains electricity (from the wall outlet) into the multiple DC voltages required by computer components. Without the PSU, nothing works — it is the heart of the system.

- Converts 110V/240V AC from the wall to regulated DC voltages: 3.3V, 5V, 12V.
- Wattage rating indicates total power delivery capacity (e.g. 650W, 850W). Insufficient wattage causes crashes or failure to boot.
- 80 Plus certification indicates efficiency rating — higher efficiency = less wasted heat.
- Connectors: main 24-pin ATX motherboard connector, 8-pin CPU connector, PCIe connectors for GPU, SATA power for drives.
- Modular PSUs allow disconnecting unused cables — cleaner builds with better airflow.
- Critical consideration: if components collectively need more power than the PSU can provide, the system will be unstable or fail to start.

Connects via: Main 24-pin connector to motherboard; dedicated connectors to CPU, GPU, and storage.

Security relevance: Physical access to cut power is a denial-of-service attack. Overloaded PSUs can damage components. UPS (Uninterruptible Power Supply) protects against unexpected power loss — sudden power cuts can corrupt file systems and databases.

🌀 GPU — Graphics Processing Unit ≈ Body: Visual Cortex

The GPU (Graphics Card) processes visual information from the OS and applications and outputs rendered images to a monitor. Modern GPUs are also used for parallel computing — machine learning, cryptomining, scientific simulation.

- Dedicated GPU: a separate PCIe card with its own VRAM (video memory) — e.g. Nvidia RTX, AMD Radeon.
- Integrated GPU: built into the CPU die — shares system RAM. Sufficient for 2D, video, and light 3D but not gaming.
- VRAM (Video RAM): high-bandwidth memory on the GPU for storing textures, frame buffers, and rendering data.
- GPU cores (CUDA cores / stream processors): thousands of small, parallel compute units — ideal for tasks that can be parallelised.
- Connects to display via HDMI, DisplayPort, or USB-C ports on the GPU.
- Modern GPUs support ray tracing and AI upscaling (DLSS/FSR) for realistic lighting and performance enhancement.

Connects via: PCIe x16 slot on the motherboard.

Security relevance: GPUs are used to accelerate password cracking (hashcat) — thousands of parallel cores process billions of hashes per second. A dedicated GPU can crack weak passwords in minutes. This is why long, complex passwords and proper hashing algorithms (bcrypt, Argon2) are essential. GPUs also accelerate cryptomining, making systems with GPUs high-value malware targets.



Input/Output (I/O) Devices ≈ Body: Senses (Eyes, Ears, Hands)

I/O devices are how humans interact with computers. Input devices feed data into the computer; output devices deliver results from the computer to the user.

- Input devices: keyboard (typed text), mouse (cursor control), microphone (audio input), webcam (video input), scanner, touchscreen.
- Output devices: monitor (visual display), speakers (audio output), printer (physical output), projector.
- Bi-directional devices: touchscreen (both input touch and output display), network interface (send and receive).
- Connect via USB (most common), PS/2 (legacy keyboard/mouse), Bluetooth (wireless), 3.5mm audio jack, HDMI/DisplayPort (display).
- USB is the universal standard — USB 2.0, 3.0, 3.1, 3.2, USB4, USB-C (connector shape).
- I/O devices communicate with the OS via device drivers — software that translates between hardware and the OS.

Connects via: USB ports, audio jacks, display ports (HDMI/DP), PS/2 ports on the motherboard rear panel.

Security relevance: Keyboard loggers (hardware and software) silently capture every keystroke. Rogue USB devices (BadUSB) emulate keyboards and inject malicious commands. Malicious charging cables contain hidden attack electronics. Physical USB port controls (port blocking, USB whitelisting) are important endpoint security measures.

3. The Boot Process — What Happens When You Press Power

The boot process is the sequence of events that occurs from the moment you press the power button to the moment the operating system is fully loaded and ready for use. Understanding the boot process is critical in security — it is a prime target for sophisticated attackers because code that runs before the OS loads can bypass OS-level security controls entirely.

Why Boot Security Matters

Bootkit and rootkit malware targets the boot process precisely because it executes before the OS and most security tools. If an attacker controls the boot process, they control everything that comes after it. UEFI Secure Boot, TPM chips, and measured boot are the defences specifically designed to protect this critical phase.

1

Press the Power Button (Body analogy: Waking up from sleep — receiving oxygen, heart starts pumping)

Pressing the power button sends an electrical signal to the PSU. The PSU begins supplying regulated power to all components — motherboard, CPU, RAM, storage, GPU. Components initialise and signal they are ready.

Security note: Physical power button is the most fundamental access control — if an attacker has physical access, they can power the system on/off at will, force cold reboots to clear RAM encryption, and attach external boot media.

2

Firmware Starts — UEFI / BIOS *(Body analogy: Body started, core systems running, but brain not yet conscious)*

The CPU fetches its first instructions from the UEFI chip on the motherboard. UEFI (Unified Extensible Firmware Interface) is the modern replacement for BIOS (Basic Input/Output System). UEFI initialises hardware at a low level — setting up memory, CPU cores, PCIe, and storage controllers. Note: BIOS and UEFI are often used interchangeably, but UEFI is the modern standard.

Security note: UEFI firmware is a prime target for nation-state-level malware (UEFI implants). Once compromised, UEFI malware survives OS reinstallation, disk replacement, and antivirus scanning — it lives on the firmware chip itself. Secure Boot validates that only signed bootloaders are allowed to execute.

3

Power-On Self Test (POST) *(Body analogy: Body running — testing all systems are functional before full consciousness)*

UEFI runs POST (Power-On Self Test) — a hardware diagnostic that tests every required component: CPU, RAM, storage, keyboard, display. If a critical component is missing or faulty, POST fails. Errors are communicated via beep codes (if a speaker is connected) or on-screen error messages.

Security note: POST errors can reveal hardware configuration details useful for attackers. Physical access during POST allows booting from external media by changing boot order. POST checks do not verify software integrity — that is the role of Secure Boot.

4

Select Boot Device *(Body analogy: Searching for the location of consciousness — finding where thoughts are stored)*

UEFI maintains a boot order list — an ordered priority of devices to check for a bootable OS (SSD, HDD, USB, network PXE). UEFI checks each device in order and loads the bootloader from the first device that responds. The boot order is configurable in the UEFI settings menu (usually accessed by pressing F2, F10, Delete, or Escape during startup).

Security note: Modifying boot order to boot from a USB drive is the standard method for: installing OS, running live security tools (Kali Linux), bypassing OS login, accessing encrypted drives. UEFI password protection and USB boot restriction are important physical security controls.

5

Bootloader Loads the OS *(Body analogy: Consciousness awakening — the brain becoming fully aware and in control)*

The bootloader (e.g. GRUB for Linux, Windows Boot Manager for Windows) is a small program at the start of the boot device. It reads the boot partition, locates the OS kernel, loads it from storage into RAM, and transfers CPU control to the OS. The OS then initialises all drivers, services, and user-space programs until the login screen appears. Once the OS is loaded, UEFI transfers control and steps back.

Security note: Bootkits replace the bootloader with malicious code — giving attackers kernel-level control before any OS security loads. UEFI Secure Boot prevents unsigned bootloaders from executing. Measured boot (TPM) records each boot stage's hash, allowing remote attestation that the system booted cleanly.

3.1 BIOS vs UEFI — Key Differences

Feature	BIOS	UEFI
Interface	Text-only, keyboard navigation	Graphical UI with mouse support
Architecture	16-bit mode only	32-bit and 64-bit modes
Disk size support	Up to 2TB (MBR partition scheme)	Up to 9 zettabytes (GPT partition scheme)
Boot method	MBR (Master Boot Record)	GPT (GUID Partition Table)
Secure Boot	Not supported	Supported — validates bootloader signatures
Network boot	Basic PXE support	Full network boot capabilities
Update method	Flash chip (risky, rare)	Firmware update via OS or USB — more accessible
Security	No hardware security features	Supports TPM, Secure Boot, measured boot
Status	Legacy — still found on older hardware	Modern standard — all current hardware uses UEFI

4. How Components Work Together

Understanding individual components is only half the picture. The real insight is how they interact during normal operation — forming the complete data processing pipeline that runs every application you use.

Scenario	Component Interaction Chain
Opening a web browser	You press Enter (I/O keyboard) → CPU reads instruction → RAM loads browser from storage → GPU renders UI to monitor (I/O display) → NIC sends HTTP request over network.
Loading a web page	NIC receives HTTP response → CPU processes data → RAM holds HTML/CSS/JS in memory → CPU parses and renders → GPU outputs rendered frame to monitor.
Saving a file	Keyboard input → CPU processes save command → RAM temporarily holds data → CPU instructs storage controller to write data → Storage writes to HDD/SSD permanently.
Running a game	Storage loads game files into RAM → CPU processes physics/AI logic → GPU renders 3D graphics (using VRAM) → Monitor displays frames → Audio outputs sound → Input devices capture controller/keyboard.

System startup

PSU powers on → CPU gets power → UEFI firmware runs → POST tests hardware → Storage loads bootloader → RAM receives OS kernel → CPU gives control to OS → All devices initialise.

5. Security Implications — The Hardware Attack Surface

Physical Access Is Total Access

If an attacker has physical access to a computer, all software-level security controls become significantly weaker. They can: remove the storage drive and read it from another machine, boot from a USB device to bypass OS authentication, reset BIOS/UEFI passwords by removing the CMOS battery, perform cold-boot attacks to extract RAM contents, and install hardware keyloggers or implants. Physical security is the foundation of all other security.

The Boot Process Is a High-Value Target

Sophisticated attackers (nation-states, APT groups) target the boot process because malware that executes before the OS is extraordinarily difficult to detect and remove. UEFI implants survive OS reinstallation and disk replacement. Secure Boot (UEFI feature) and TPM-based measured boot are the primary defences — they ensure only signed, verified code executes at boot time.

RAM Contains Secrets — Memory Forensics

Because RAM holds all active data — including encryption keys, authentication tokens, plaintext credentials, and decrypted documents — a running system's RAM is a goldmine for attackers and forensic investigators alike. Tools like Volatility analyse RAM dumps to find malware processes, injected code, network connections, and credentials. Cold-boot attacks exploit RAM's brief persistence after power loss.

Storage Is the Evidence — And the Target

All persistent data lives on storage — documents, OS files, application data, logs, browser history. Ransomware targets storage by encrypting files and demanding payment. Digital forensics recovers deleted files from storage. Full-disk encryption (FDE) protects data at rest — if a device is stolen, FDE prevents the data from being read without the decryption key, even if the drive is removed.

GPUs Accelerate Password Cracking

Modern password cracking tools (hashcat) leverage GPU parallelism to test billions of password candidates per second. A modern gaming GPU can crack an MD5 hash of a short password in milliseconds. This is why modern password storage must use slow hashing algorithms (bcrypt, Argon2, scrypt) that are computationally expensive by design — making GPU acceleration significantly less effective.

USB and I/O Devices Are Covert Attack Vectors

USB devices can impersonate keyboards (BadUSB/Rubber Ducky attacks) and inject keystrokes silently when plugged in. Hardware keyloggers between keyboard and USB port capture every keystroke invisibly. Malicious charging cables contain attack electronics. Endpoint security must include USB device control, disabling unused ports, and user awareness training about plugging in unknown devices.

6. Key Takeaways & Glossary

6.1 Key Takeaways

- Every computer — desktop, laptop, server, smartphone — is built from the same fundamental components.
- The human body analogy: Motherboard = Skeleton/Nerves, CPU = Brain, RAM = Short-term memory, Storage = Long-term memory, NIC = Vocal cords, PSU = Heart, GPU = Visual cortex, I/O = Senses.
- RAM is volatile — loses all data on power loss. Storage is non-volatile — retains data without power.
- HDDs use spinning magnetic platters (mechanical, slower, cheaper per GB). SSDs use flash chips (no moving parts, faster, more durable, more expensive per GB).
- The 5-step boot process: Power On → UEFI/BIOS → POST → Select Boot Device → Bootloader loads OS.
- UEFI is the modern replacement for BIOS — adds Secure Boot, GPT support, 64-bit mode, and hardware security features.
- POST (Power-On Self Test) checks all hardware is present and functional before loading the OS.
- Physical access = highest privilege level — physical security controls are the foundation of all security.
- The boot process is targeted by sophisticated malware (UEFI implants, bootkits) because pre-OS code bypasses OS security.
- GPU parallel processing is used by attackers for password cracking — use slow hashing algorithms (bcrypt, Argon2).

6.2 Core Glossary

Term	Definition
Motherboard	The main circuit board connecting all components — skeleton and nervous system of the computer.
CPU	Central Processing Unit — executes all program instructions. Brain of the computer.
Core	An independent processing unit within a CPU. Modern CPUs have 4–24+ cores for parallel execution.

Clock Speed	CPU cycles per second (GHz) — how many instruction sets the CPU can process each second.
RAM	Random Access Memory — fast volatile working memory. Cleared on power loss.
DDR5	Double Data Rate 5 — current generation RAM technology offering high speeds and efficiency.
Volatile	Data is lost when power is removed. RAM is volatile; storage is non-volatile.
HDD	Hard Disk Drive — mechanical storage using spinning magnetic platters. Slower, higher capacity per cost.
SSD	Solid State Drive — flash memory storage. No moving parts, much faster, more durable than HDD.
NVMe	Non-Volatile Memory Express — high-speed SSD interface using PCIe, much faster than SATA.
SATA	Serial ATA — interface for connecting HDDs and SATA SSDs to the motherboard.
PCIe	Peripheral Component Interconnect Express — high-bandwidth slot for GPU, NVMe SSD, and NIC cards.
NIC	Network Interface Card — hardware enabling wired or wireless network communication. Vocal cords analogy.
PSU	Power Supply Unit — converts AC mains power to DC voltages for all computer components. Heart analogy.
GPU	Graphics Processing Unit — renders visual output to display. Also used for parallel computing tasks.
VRAM	Video RAM — high-bandwidth memory on the GPU for textures and frame buffers.
UEFI	Unified Extensible Firmware Interface — modern firmware replacing BIOS. Runs at startup before OS loads.
BIOS	Basic Input/Output System — legacy firmware largely replaced by UEFI. Often used interchangeably.
POST	Power-On Self Test — UEFI routine that tests all hardware before the OS loads.
Boot Order	UEFI priority list of devices to check for a bootable OS (SSD, HDD, USB, network).
Bootloader	Small program on boot device that loads the OS kernel into RAM (e.g. GRUB, Windows Boot Manager).
Secure Boot	UEFI feature that only allows signed, verified bootloaders to execute — prevents bootkit malware.

TPM	Trusted Platform Module — security chip for key storage, measured boot, and device attestation.
Cold Boot Attack	Extracting data from RAM shortly after power loss — RAM contents persist briefly when chilled.
BadUSB	Attack where a USB device emulates a keyboard and injects malicious keystrokes when plugged in.
Memory Forensics	Analysing a RAM dump (e.g. with Volatility) to find running processes, malware, and credentials.
Full Disk Encryption	Encrypts entire storage drive — protects data even if drive is physically removed (e.g. BitLocker).

6.3 Next Steps on TryHackMe

- **Immediate:** Continue with the Computer Types room — how components are combined for different device types (servers, mobile, embedded).
- **Foundation:** Study the Windows Fundamentals rooms — how the OS layers above the hardware covered here.
- **Foundation:** Complete the Linux Fundamentals rooms — the OS most used in servers and security tools.
- **Security:** Learn about memory forensics with Volatility — apply RAM knowledge for malware analysis.
- **Advanced:** Study the Malware Analysis rooms — understand how malware uses hardware (RAM, storage, NIC) to persist and spread.

Hardware Is the Root of Trust

Every security control — encryption, authentication, logging, firewalls — ultimately runs on hardware. Understanding hardware means understanding the foundational layer of all computing. The components and boot process covered in this room appear constantly in advanced security topics: memory forensics, firmware analysis, hardware security modules, and endpoint detection. This knowledge is not just background — it is foundational.

Source: TryHackMe — Inside a Computer Room | tryhackme.com/room/insideacomputer